

Parsing XML with *xml.el* in Emacs Lisp

The *xml.el* GNU Emacs Lisp file provides useful functions to process XML files. In this article, the ninth one in the GNU Emacs series, we shall explore the APIs provided to parse XML files.

The code is released under the GNU General Public License v3.0, and is included as part of GNU Emacs. It was written by Emmanuel Briot, and is now maintained by Mark A. Hershberger. From any GNU Emacs buffer, you can simply *require* the library, as shown below:

```
(require 'xml)
```

If you press *C-x C-e* at the end of the above S-expression, it should simply return *xml* in the mini-buffer. This means that the library has been loaded. The entire source code is available at <https://github.com/emacs-mirror/emacs/blob/master/lisp/xml.el>.

Examples

Let us go through the various functions, both external and internal, that are made available in the library, with examples.

Consider the following input *sample.xml* file saved in the */tmp* directory:

```
<book id="compiler">
  <bookinfo>
    <bookbiblio>
      <title>My own book!</title>
      <edition>First</edition>
      <authorgroup>
        <author>
          <firstname>John</firstname>
          <surname>Wiegley</surname>
        </author>
      </authorgroup>
    </bookbiblio>
  </bookinfo>
  <chapter>
    <title>Chapter 1</title>
    <para>A B C</para>
  </chapter>
</book>
```

The file contents can be inserted into a temporary buffer, and the *xml-parse-region* function can be executed to parse the XML, and store the result to a *doc* variable as shown below:

```
(xml-parse-region &optional beg end buffer parse-dtd parse-ns) ;; Syntax
```

```
(setq doc (with-temp-buffer
            (insert-file-contents "/tmp/sample.xml")
            (xml-parse-region (point-min) (point-max))))
((book ((id . "compiler")) "
  (bookinfo nil "
    (bookbiblio nil "
      ... "
      ... "
      ... "
    ) "
  ) "
  (chapter nil "
    (title nil "Chapter 1") "
    (para nil "A B C") "
  ) "
))
```

The *point-min* function returns the minimum permissible value of the point in the current buffer, which is 1. The *point-max* function returns the maximum permissible value of point in the current buffer, which is *1 + buffer-size*. The source code of the *xml-parse-function* is as follows:

```
(defun xml-parse-region (&optional beg end buffer parse-dtd
  parse-ns)
  (unless buffer
    (setq buffer (current-buffer)))
  (with-temp-buffer
    (insert-buffer-substring-no-properties buffer beg end)
    (xml--parse-buffer parse-dtd parse-ns)))
```